

Task 1 Report: SOMLAP Image Exploration and Related Work

Md. Rafiqul Islam (2026-2-74-025)
Ahanaf Shahadat Taseen (2026-2-74-033)
Reasad Mahmood (2026-2-74-012)

Department of Computer Science and Engineering, East West University

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1. Dataset provenance and problem definition

SOMLAP was released for static detection of Windows malware from Portable Executable (PE) metadata [2]. The source article states that the dataset contains 51,409 files and 108 header-derived attributes. The supplied CSV has 51,408 records: 31,600 benign files and 19,808 malware files. One malware record is therefore absent from the local copy. To keep every calculation reproducible, the analysis uses the observed file rather than adjusting its counts to match the article.

The response variable, *class*, assigns label 0 to benign software and label 1 to malware. All 108 predictors are numeric and describe PE headers or sections, including sizes, virtual addresses, entropy, version fields and section characteristics. The file contains no raw executable bytes and no malware-family labels. The resulting problem is binary classification from static tabular evidence.

For the image experiment, the 108 values are placed on a fixed 12×9 grid in CSV column order. Resizing this grid permits convolution and attention without dropping a predictor, but it also imposes a neighborhood structure that is not part of the PE format. Consequently, image-based explanations are read as sensitivity to feature positions. They are not treated as localization of malicious instructions or executable regions.

2. Image EDA protocol

2.1 Summary, labels and data types

The audit records the file dimensions, schema, data types, descriptive statistics and label values before any transformation. The table has 109 columns: 108 predictors and one binary target. Establishing this schema at the outset prevents later preprocessing steps from silently adding, removing or reordering fields.

2.2 Class balance

Benign records form 61.47% of the sample and malware records 38.53%, a ratio of approximately 1.60:1. The minority class is large enough to support direct estimation of its error rates, but the imbalance can still make accuracy look favorable when malware recall is weak. Later experiments therefore use Macro-F1 as the main selection criterion and report malware precision, recall and F1 alongside accuracy.

Class	Label	Count	Share
Benign	0	31,600	61.47%
Malware	1	19,808	38.53%
Total	-	51,408	100.00%

Table 1. Class distribution in the supplied SOMLAP CSV.

2.3 Missing values, duplicates and sentinel values

No column contains a missing value, although completeness alone does not establish data integrity. The CSV includes 14,367 exact duplicate rows, or 27.95% of all observations. With an ordinary row-wise split, copies of the same 108-value vector could occur in training and testing, allowing memorization to inflate performance. The modeling protocol therefore hashes the full feature vector and assigns all copies of a hash to one partition.

Thirty-six predictors meet the notebook’s 99% near-constant criterion. Six fields contain 4,294,967,295, the largest unsigned 32-bit integer. Depending on the original parser, that value may represent a legitimate flag, a sentinel or an extraction artifact. It is retained because the available documentation does not justify recoding it. Its presence, together with the near-constant fields, is carried forward as a limitation of the source data.

2.4 Distribution, correlation and low-dimensional structure

The univariate distributions contain heavy right tails, repeated zeros and isolated extreme values. Several predictors show visible class shifts, whereas others overlap almost completely. Correlation analysis also identifies groups of related header and section fields. These dependencies can supply useful combinations to a classifier, but their placement next to one another in the grid depends on column order rather than a domain-defined topology.

Principal-component projections show partial, not complete, separation of the two classes. Considerable overlap remains in the first components, so the EDA does not support a claim that a simple linear boundary solves the task. It does show that class information is distributed across several directions, which motivates evaluating nonlinear models under controlled splits.

3. EDA of the generated images

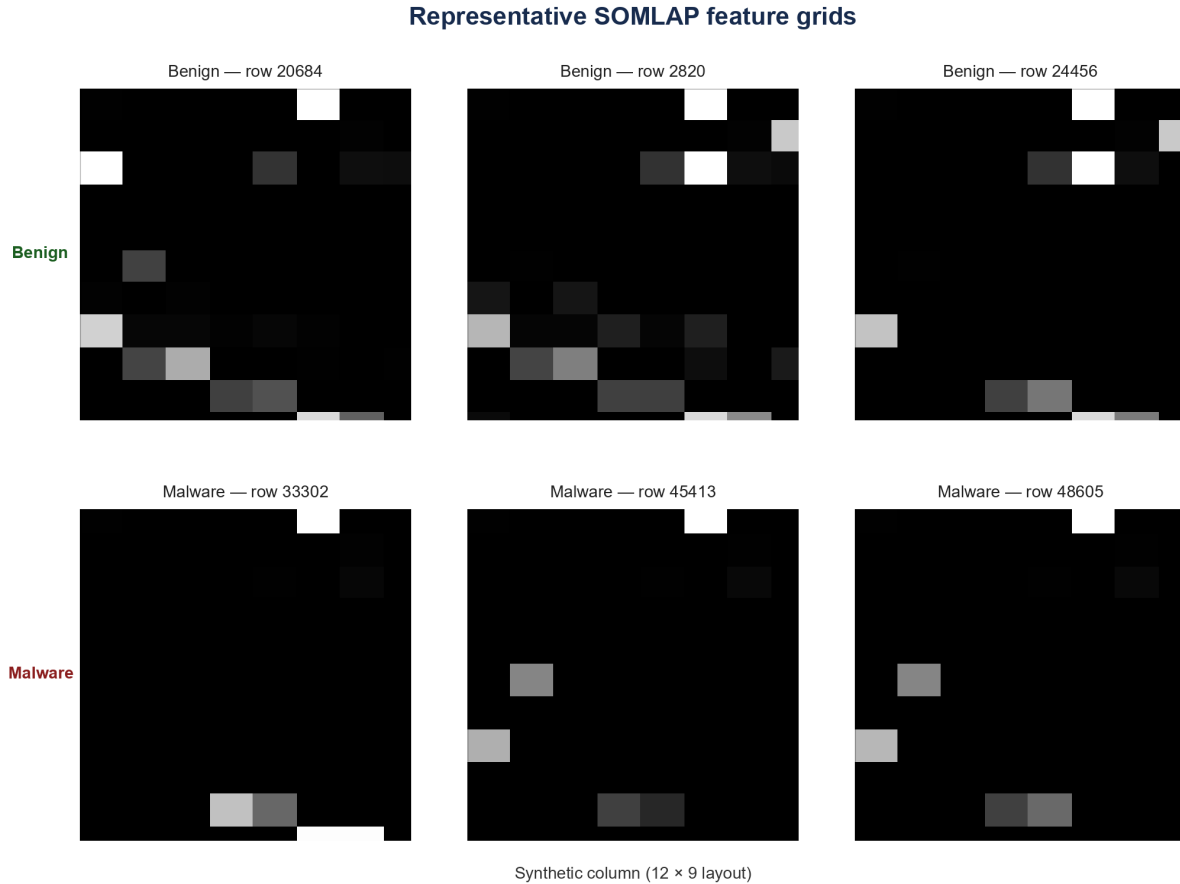


Figure 1: Six labelled SOMLAP feature grids after descriptive min-max scaling.

Figure 1 presents three benign and three malware records selected with seed 42. Bright cells reflect large scaled feature values at particular column positions. Apparent blocks or textures arise from the numeric values and their ordering; they do not correspond to sections of executable code.

3.1 Size, resolution and format

Every record begins as a 12×9 single-channel grid and enters the model as a $96 \times 96 \times 3$ array. The spatial dimensions are enlarged by interpolation, and the grayscale plane is copied into three channels to satisfy the interfaces of transfer-learning networks. Width, height, aspect ratio and color therefore carry no independent sample information. Conventional checks for camera blur, compression artifacts or exposure are not applicable.

3.2 Pixel distribution and numeric quality

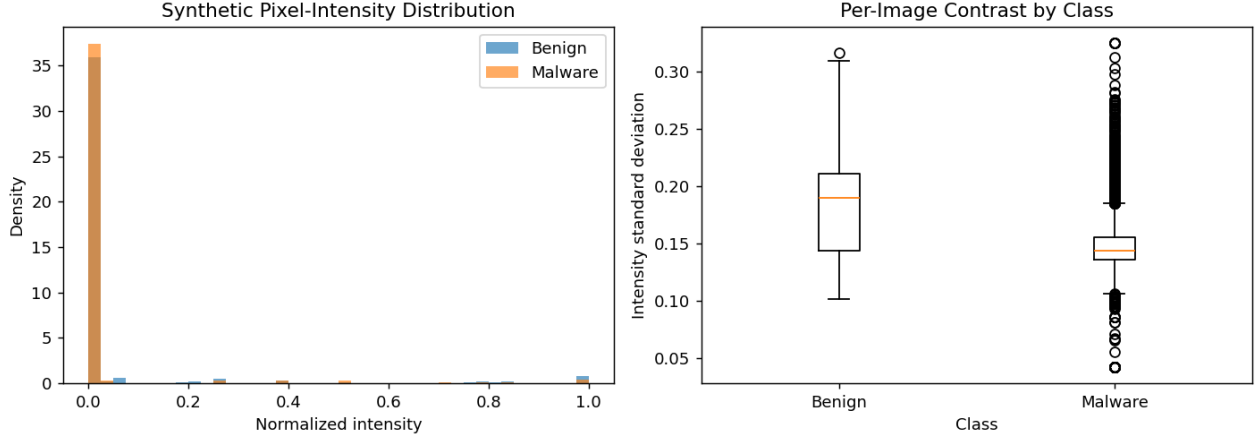


Figure 2: Normalized pixel density and within-image contrast by class.

Descriptive min-max scaling produces finite pixels throughout the collection. Zero accounts for 57.47% of grid positions and one for 1.51%; the median within-grid standard deviation is 0.1564. The mass at zero is consistent with sparse or inactive PE fields. Malware records have a narrower central contrast distribution in this sample, although the class tails overlap and contrast alone cannot separate them.

Copying the grayscale plane into RGB does not add information. It only makes the data compatible with VGG16, ResNet50 and EfficientNetB0. Any advantage of those networks must therefore come from their learned operations, not from new color channels introduced during conversion.

4. Related work and research gap

The literature review combines five recent attention-based malware-image studies with two papers that evaluate SOMLAP directly. This distinction is important: the first group informs attention design, while the second provides dataset-specific performance context. None supplies the exact partition or per-sample predictions needed for a paired test against the present results.

Study	Year	Dataset	Method	Attention	Reported result	Relevance to this study
Wang et al.	2022	MalImg; BIG 2015	CliqueNet with MSAAM	Hybrid channel and spatial	Detection 99.8%; family accuracy 99.2% and 98.2%	Supplies the channel-spatial design rationale

Study	Year	Dataset	Method	Attention	Reported result	Relevance to this study
Kattamuri et al.	2023	SOMLAP	ACO, CSO and GWO feature selection with classical ML	None	Best accuracy 99.37% using ACO and 12 features	Dataset source and contextual benchmark
Ravi and Alazab	2023	MalImg-style images, 25 families	CNN with multi-head attention	Multi-head self-attention	About 99% accuracy	Supports a controlled attention experiment
Huang et al.	2023	MalImg; Blended+	CoAtNet with depthwise separable convolutions	Self-attention with convolution	99.33% and 96.60% accuracy	Provides a convolution-attention comparison point
Fan et al.	2025	MalImg; BIG 2015	ResNet with global channel-spatial attention	Channel and spatial	99.08% and 98.58% accuracy	Closest recent channel-spatial architecture
Liu et al.	2026	MaleVis	AlexNet transfer learning with MHSA	Multi-head self-attention	87.3% accuracy	Shows a heavier attention option
Alwateer et al.	2026	SOMLAP and other PE datasets	Hierarchical deep network with DRBCE loss	None stated	SOMLAP accuracy 99.11%; weighted F1 98.85%	Strongest verified SOMLAP context; not a paired comparison

Table 2. Related work used to define the research question. Full limitations and research gaps appear in the separate related-work table.

Wang et al. combine channel and spatial attention in MSAAM and report 99.2% family accuracy on MalImg [1]. Ravi and Alazab evaluate multi-head attention in a CNN [3], Huang et al. combine depthwise convolution with self-attention [4], and Fan et al. place global channel-spatial attention inside a residual network [5]. Liu et al. add multi-head self-attention to AlexNet and include Grad-CAM examples [6]. Their results establish that attention can be useful on byte-derived malware images, but they do not show that the same conclusion holds when tabular PE fields are placed on a fixed grid.

For SOMLAP, Kattamuri et al. report 99.37% accuracy after ACO selects 12 attributes [2]. Alwateer et al. report 99.11% accuracy and 98.85% weighted F1 with a hierarchical network [7]. These figures are not protocol-matched benchmarks: the first changes the input representation, and neither paper publishes the split assignments or sample-level predictions.

The unresolved question is whether CBAM changes balanced classification when the 108-field representation, duplicate-safe partition, backbone and training conditions are held constant. A related interpretive question is whether local explanations can be tied to named feature slots without presenting the grid as a physical malware image. The subsequent experiments address these questions with matched controls, grouped folds and two explanation methods.

5. Modeling decisions

The EDA leads to five experimental controls:

- Exact feature vectors define groups, and every group remains within a single training, validation or test partition.

- Scaling is estimated from training data and then applied unchanged to validation and test records.
- Class weights are used during fitting, while Macro-F1 and malware recall remain central to model comparison.
- Augmentation is confined to training; validation and test pipelines are deterministic.
- Grad-CAM and LIME are interpreted over synthetic feature positions, not as causal evidence or localization of malicious bytes.

6. Limitations of the exploratory evidence

Grouping exact feature vectors prevents direct duplicate leakage, but it cannot separate related variants from the same malware family or collection source. The supplied CSV contains no stable family, executable hash, host or timestamp identifier from which a stronger source-wise split could be constructed. Near-duplicates may therefore remain across partitions.

The image representation introduces a separate construct limitation. Its 12×9 neighborhood follows CSV order, not an executable layout or a learned feature topology. In addition, the available file is one record smaller than the source description and no external or temporal cohort is supplied. The reports retain these boundaries rather than generalizing the results beyond the analyzed file.

References

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